KENYA CERTIFICATE OF SECONDARY EDUCATION

COMPUTER PROJECT 2026 (451/3)

NAME:…………………………………….

SCHOOL CODE……………………………….

SCHOOL……………………………………

INDEX NUMBER…………………………………

PROJECT TITLE: AZANI INTERNET SERVICE INFORMATION PROVIDER SYSTEM

YEAR OF EXAMINATION:2026

DECLARATION

STUDENT'S DECLARATION

I, …………………………., declare that this Computer Studies Project on the AZANI internet service provider information System is my original work. All the information, and documentation presented in this project are authentic and have been developed solely for the purpose of fulfilling the requirements of the Computer Studies examination. Any external sources referenced in this project have been appropriately cited. I understand the consequences of academic dishonesty, and I affirm that this project represents my independent efforts and knowledge acquired during the course. I take full responsibility for the content presented.

Signature: ___________________________

Date: ___________________________

SUPERVISOR DECLARATION

I,……………….., hereby declare that I have supervised and guided Emmanuel Kimaru in the development of the AZANI internet service provider information System for the Computer Studies Project. I have reviewed the project documentation, and implementation, and I affirm that the work presented is consistent with the guidelines and expectations of the KNEC. I believe that the student has demonstrated a sound understanding of the concepts and skills covered in the Computer Studies curriculum. I take responsibility for providing necessary guidance and support throughout the project's development.

Signature: ___________________________

Date: ___________________________

DEDICATION[i]

I dedicate this Computer Studies Project on the to my Parents. Your unwavering support, encouragement, and guidance have been instrumental throughout this academic journey. This project is a reflection of the inspiration and motivation drawn from your belief in my abilities. I express my deepest gratitude for the sacrifices made and the encouragement given. This work stands as a tribute to your influence on my educational pursuits.

ACKNOWLEDGMENT

I would like to express my sincere appreciation to everyone who contributed to the successful completion of this Computer Studies Project on the AZANI Internet service provider information System. My heartfelt gratitude goes to my supervisor, Trainer, for their invaluable guidance, constructive feedback, and unwavering support throughout the project. I am also grateful to my mentors for providing the necessary knowledge and inspiration that fueled my understanding of the subject matter. Special thanks to my school for their encouragement and assistance during challenging moments. Lastly, I extend my appreciation to my family and friends for their understanding, patience, and encouragement throughout this academic endeavor.

Contents

CHAPTER ONE: INTRODUCTION

BACKGROUND INFORMATION

Azani is a company that deals with provision of internet services and internet infrastructure to learning institutions. These institutions include primary, junior, senior schools and colleges. Institutions interested in the services are required to pay a registration fee of Ksh 8,500 to the company.

When an institution requests for their services, the company visits the site to establish the readiness of the institution to have internet installed. The company finds out the number of users and infrastructure requirements such as computers and Local Area Network (LAN). An institution that has all the requirements and is ready for connectivity, pays an installation fee of Ksh 10,000. If the institution has not met the requirements, they may purchase personal computers from Azani Company at Ksh 40,000 each

An institution that would like to upgrade to a higher bandwidth is offered a 10% discount of the cost of the bandwidth they are upgrading to.
All institutions using the internet services are required to pay by the end of the current month. An institution which fails to pay is surcharged a fine of 15% of the amount as an overdue fine. The internet services will be disconnected if the amount due and the fine will not have been paid by the 10th day of the subsequent month. Once an institution pays the monthly bill and the overdue fine, a fee of Ksh 1,000 is surcharged for re-connection

SYSTEM ANALYSIS

1.1 PROBLEM DEFINITION

Internet Service Providers (ISPs) such as Azani face challenges in managing customer information, billing, and service requests efficiently. The current methods of record-keeping are largely manual or rely on basic spreadsheets, which often lead to errors, duplication of data, delayed service delivery, and difficulties in generating accurate reports. Customers may experience delays in registration, billing disputes, and poor tracking of service usage.

There is therefore a need for a computerized information system that can streamline customer registration, automate billing, track payments, and generate timely reports. Such a system will improve accuracy, reduce redundancy, enhance customer satisfaction, and support better decision-making for the ISP.

1.2 OVERVIEW OF THE EXISTING SYSTEM

Currently, Azani Internet Service Provider relies on manual and semi-computerized methods to manage customer information and service delivery. Customer registration is often recorded in physical files or basic spreadsheets, which makes retrieval of data slow and prone to errors. Billing is handled through manual invoice preparation or simple word-processing templates, leading to delays and inconsistencies in payment tracking.

Service requests and complaints are logged in notebooks or through phone calls without a centralized tracking system, making it difficult to monitor resolution progress. Reports such as customer lists, payment summaries, and service usage statistics are generated manually, consuming significant time and resources. The lack of integration between different records results in duplication, inefficiency, and poor customer service.

1.2.1 SYSTEM STRUCTURE

The current system at AZANI School lacks a centralized and integrated structure, relying on disparate manual processes that contribute to operational inefficiencies. Club information is stored in physical files, making it difficult to maintain a unified and easily accessible . Each service category, including membership, financial contribution's, and savings, operates independently, leading to challenges in cross-referencing and retrieving relevant data. The absence of a standardized data model results in redundant data entry and increases the likelihood of errors. Furthermore, the manual computation of fees and charges lacks a systematic approach, often causing delays and inaccuracies in financial transactions. The absence of a structured system architecture not only impedes the efficiency of day-to-day operations but also hinders the school's ability to adapt to evolving business needs. The proposed computerized system aims to rectify these structural deficiencies by providing a centralized, organized, and interconnected platform that streamlines data management, service computations, and financial tracking for AZANI School club membership system..

1.3 OVERVIEW OF THE PROPOSED SYSTEM

The proposed Azani Internet Service Provider Information System will be a computerized solution designed to automate customer management, billing, and service tracking. It will consist of a relational database that stores customer details, services subscribed, and payment records. The system will provide user-friendly input forms for registering new customers, updating service requests, and recording payments.

Billing will be automated through queries that calculate charges based on subscribed services and payment history, reducing errors and delays. Reports such as invoices, receipts, customer lists, and payment summaries will be generated instantly, improving efficiency and accuracy. The system will also include search and filter functions to quickly retrieve customer information and monitor outstanding balances.

By integrating all records into a single database, the system will eliminate duplication, enhance data security, and support faster decision-making. Overall, the proposed system will improve customer satisfaction, streamline operations, and provide reliable information for management

1.3.1 OBJECTIVES OF THE PROPOSED SYSTEM

Objectives of the Proposed Azani ISP Information System

1. **Automate Customer Registration**
    - To provide a digital platform where customer details can be entered, validated, and stored in a centralized database, reducing errors and duplication.
2. **Streamline Service Subscription Management**
    - To record and track services subscribed by each customer, ensuring accurate billing and easy monitoring of service usage.
3. **Automate Billing and Payment Tracking**
    - To generate invoices automatically, record payments, and update balances in real time, minimizing disputes and delays.
4. **Generate Accurate Reports**
    - To produce timely reports such as invoices, receipts, customer lists, and payment summaries, supporting management decisions and customer communication.
5. **Enhance Data Security and Integrity**
    - To safeguard customer and financial records through controlled access and proper database management, ensuring reliability of information.
6. **Improve Customer Service**
    - To reduce waiting times, provide clear billing information, and enable faster resolution of service requests, thereby increasing customer satisfaction.
7. **Support Decision-Making**-To provide management with accurate, up-to-date information on customer trends, payments, and service usage for strategic planning
    -

1.3.2 BENEFITS OF THE PROPOSED SYSTEM

Benefits of the Proposed System

1. **Improved Accuracy**
   - Automated data entry and validation reduce errors in customer records, billing, and payment tracking.
2. **Time Efficiency**
   - Tasks such as invoice generation, report preparation, and customer searches are completed faster compared to manual methods.
3. **Better Customer Service**
   - Customers receive timely invoices, receipts, and faster responses to service requests, enhancing satisfaction.
4. **Centralized Data Management**
   - All customer, service, and payment records are stored in one database, eliminating duplication and making retrieval easier.

1.3.3 Disadvantages of the Proposed System

1. **High Initial Cost**
   - Developing and setting up the system requires investment in software, hardware, and training, which may be expensive for the ISP.
2. **Technical Expertise Required**
   - Staff must be trained to use and maintain the system. Without proper skills, errors or misuse may occur.
3. **Dependence on Electricity and Internet**
   - The system relies on stable power and internet connectivity. Outages can disrupt operations and access to records.
4. **Risk of Data Loss**
   - If backups are not properly maintained, system crashes or hardware failures could lead to loss of important customer and financial data.

5. **Security Threats**
    - The system may be vulnerable to hacking, unauthorized access, or malware if strong security measures are not implemented.
6. **Resistance to Change**
    - Employees accustomed to manual methods may resist adopting the new system, slowing down implementation.
7. **Maintenance Costs**

- Regular updates, repairs, and technical support add to the long-term expenses of running the system.

1.3.4 SCOPE OF THE SYSTEM

Customer Registration

The system will capture and store customer details such as name, contact information, and service subscription in a centralized database.

Service Subscription Management

It will record services subscribed by each customer, including package type, duration, and charges.

Billing and Payment Tracking

The system will generate invoices automatically, record payments made, and update outstanding balances.

Report Generation

It will produce reports such as customer lists, payment summaries, invoices, and receipts for management and customer use.

1.3.5 COST AND BENEFITS ANALYSIS

Costs

1. **Initial Development Costs**
   - Expenses for software tools (e.g., database management system, programming environment).
   - Hardware requirements such as computers and storage devices.
2. **Training Costs**
   - Time and resources needed to train staff on how to use and maintain the system.
3. **Maintenance Costs**
   - Regular updates, troubleshooting, and technical support to keep the system running smoothly.
4. **Data Backup and Security Costs**

- Investment in backup solutions and security measures to protect customer and financial data.

Benefits

1. **Improved Accuracy**
   - Automated processes reduce errors in billing, payments, and customer records.
2. **Time Efficiency**
   - Faster generation of invoices, reports, and customer searches compared to manual systems.
3. **Enhanced Customer Service**
   - Customers receive timely invoices, receipts, and faster responses to service requests.
4. **Centralized Data Management**
   - All records are stored in one database, eliminating duplication and making retrieval easier.
5. **Better Decision-Making**

- Accurate reports and summaries provide management with reliable information for planning.

6. **Cost Savings in the Long Term**

- Reduced paperwork, minimized manual labor, and streamlined operations lower administrative costs.

1.3.6 FEASIBILITY STUDY

1. **Technical Feasibility**

- The system can be developed using readily available software tools such as Microsoft Access, MySQL, or other database management systems.
- Hardware requirements (computers, printers, storage devices) are affordable and accessible to the ISP.
- Staff can be trained to use the system with minimal technical challenges.
- Therefore, the project is technically feasible because the required resources and skills are available.

2. **Economic Feasibility**

The feasibility study serves as a critical phase in the evaluation and planning of the proposed computerized system for AZANI company . This study aims to systematically assess the viability and practicality of implementing the system within the organizational context. It involves a comprehensive analysis of technical, economic, operational, and scheduling aspects to determine whether the proposed system aligns with the strategic goals of AZANI company and offers a favorable return on investment. By delving into the feasibility aspects, this study provides key insights that will guide decision-makers in determining the project's viability and in formulating informed choices throughout the system development lifecycle.

3. **Operational Feasibility**

- The system will be easy to use, with simple input forms and clear output reports.
- Staff will adapt quickly after training, and customers will benefit from faster service delivery.
- The system integrates well into existing ISP operations without disrupting core services.
- Therefore, the project is operationally feasible.

4. **Schedule Feasibility**

- The project milestones (System Analysis, Design, Construction, Documentation) are spread across several months as per the KCSE assessment sheet.
- With proper planning and adherence to timelines, the project can be completed within the allocated period.
- This makes the project schedule realistic and achievable.

1.3.8 FACT FINDING

To develop the Azani ISP Information System, several fact-finding techniques were used to collect information about the existing system and its challenges:

1. **Interviews**

- Conducted discussions with ISP staff to understand how customer registration, billing, and service requests are currently handled.
- Gathered insights into difficulties faced, such as delays in invoice preparation and tracking payments.

2. **Observation**

- Observed the daily operations of the ISP, including how customer details are recorded and how payments are processed.
- Noted inefficiencies such as duplication of records and reliance on manual files.

3. **Document Review**

- Examined existing records such as customer files,
- Reviewed literature and online resources on ISP management systems

4. **Questionnaires**

- Distributed simple questionnaires to staff and customers to collect feedback on the current system.
- Responses highlighted issues like poor customer service, delayed billing, and difficulty in retrieving information.

5. **Research**

- Reviewed literature and online resources on ISP management systems.
- Compared best practices with the current system to identify areas for improvement

1.3.9 SYSTEM REQUIREMENTS AND SPECIFICATIONS

1. Hardware Requirements

- **Processor:** At least Intel Core i3 or equivalent.
- **RAM:** Minimum 4 GB for smooth operation.
- **Storage:** At least 250 GB HDD/SSD to store customer records and backups.
- **Monitor:** Standard display for user interface interaction.
- **Printer:** For printing invoices, receipts, and reports.

2. **Software Requirements**

- **Operating System:** Windows 10 or later.
- **Database Management System:** Microsoft Access / MySQL.
- **Programming Environment:** Visual Basic, Java, or any suitable front-end tool.
- **Office Tools:** MS Word/Excel for documentation and reporting.
- **Security Software:** Antivirus and backup utilities to protect data

3. **Functional Requirements**

- Ability to register new customers and update their details.
- Ability to record service subscriptions and manage their status.
- Automated billing and payment tracking.
- Generation of invoices, receipts, and financial reports.
- Search and filter functions for quick data retrieval.
- User access control (administrator vs. staff users).

4. **Non-Functional Requirements**

- **Usability:** Simple and user-friendly interface for staff.
- **Reliability:** Accurate data processing with minimal errors.
- **Security:** Controlled access and data protection against unauthorized use.
- **Performance:** Fast response time for queries and report generation.
- **Scalability:** Ability to expand the system as the ISP grows.
- **Maintainability:** Easy to update and maintain with minimal downtime.

5. **System Specifications**

- Centralized relational database with linked tables (Customers, Services, Payments, Subscriptions).
- Forms for input (customer registration, service subscription, payment entry).
- Reports for output (customer lists, invoices, receipts, payment summaries).
- Queries for data manipulation (billing, overdue payments, service usage).
- Documentation including user manual and project report.

SYSTEM DESIGN

2.1 SYSTEM FLOWCHARTS

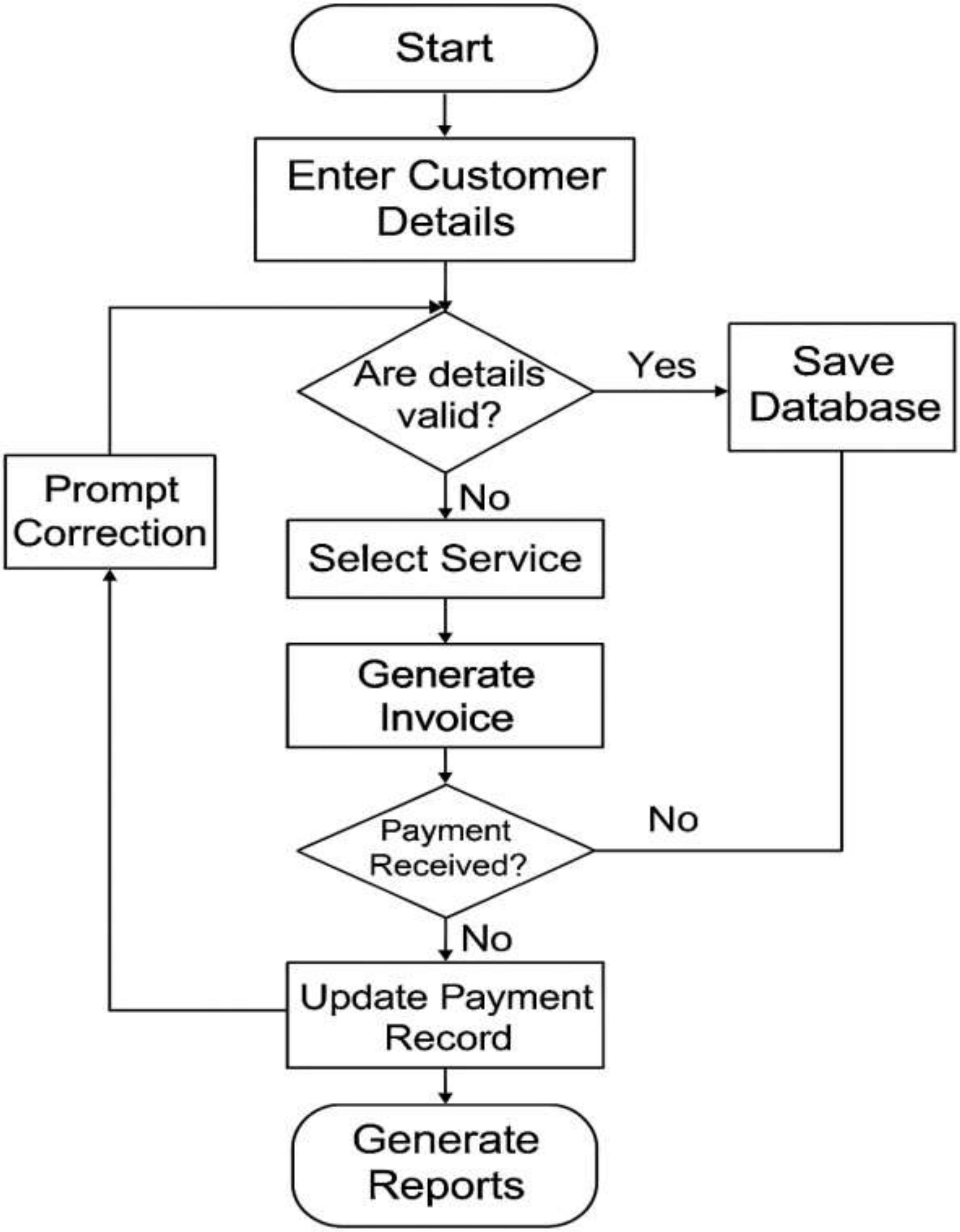


Figure 1:Registration flowchart

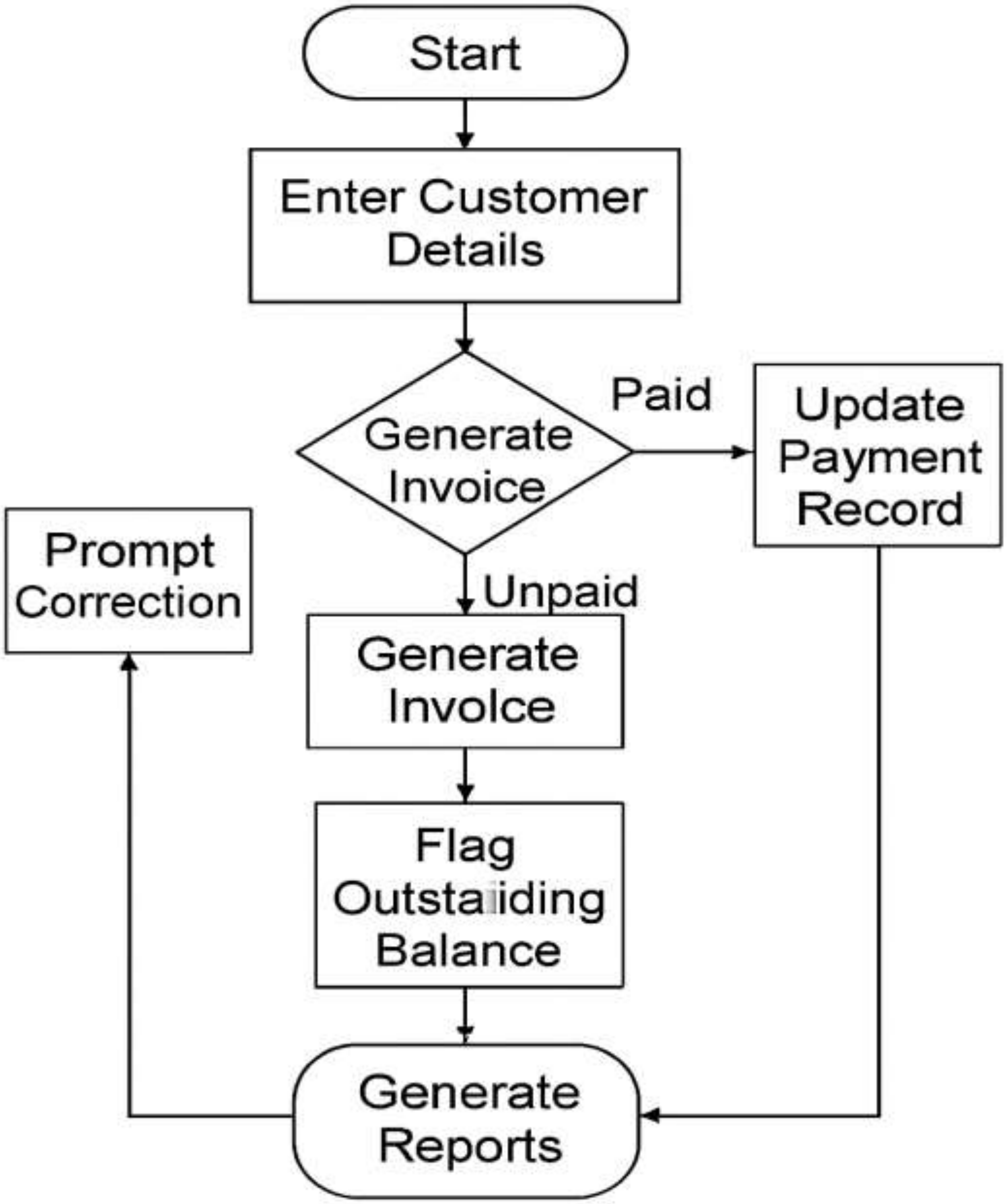


Figure 2: Payment flowchart

2.2 Fields and Data types

The proposed Azani ISP Information System will use a relational database with several tables. Each table will contain specific fields with appropriate data types to ensure accuracy and consistency.

TABLES

a. Institution

Field Name	Data Type
InstitutionID	AutoNumber (PK)
Name	Short Text
Type	Short Text (or Lookup)
Address	Short Text
ContactPersonName	Short Text
ContactPhone	Short Text
ContactEmail	Short Text

Table 1 Institution

b. Service

Field Name	Data Type
ServiceID	AutoNumber (PK)
Bandwidth	Short Text
MonthlyCost	Currency

Table 2 Service

c. Infrastructure

Field Name	Data Type
InfraID	AutoNumber (PK)
InstitutionID	Number (FK)
ComputersPurchased	Number
LANNodesPurchased	Number
LANCost	Currency
ComputerCost	Currency

Table 3 infrastructure

d. Payment

Field Name	Data Type
PaymentID	AutoNumber (PK)
InstitutionID	Number (FK)
PaymentType	Short Text (or Lookup)
Amount	Currency
DatePaid	Date/Time
Status	Short Text (or Lookup)

Table 4 Payment

e. ConnectionStatus

Field Name	Data Type
StatusID	AutoNumber (PK)
InstitutionID	Number (FK)
ServiceID	Number (FK)
CurrentBandwidth	Short Text
IsConnected	Yes/No
LastPaymentDate	Date/Time
DisconnectionDate	Date/Time

Table 5 Connection status

ER DIAGRAM

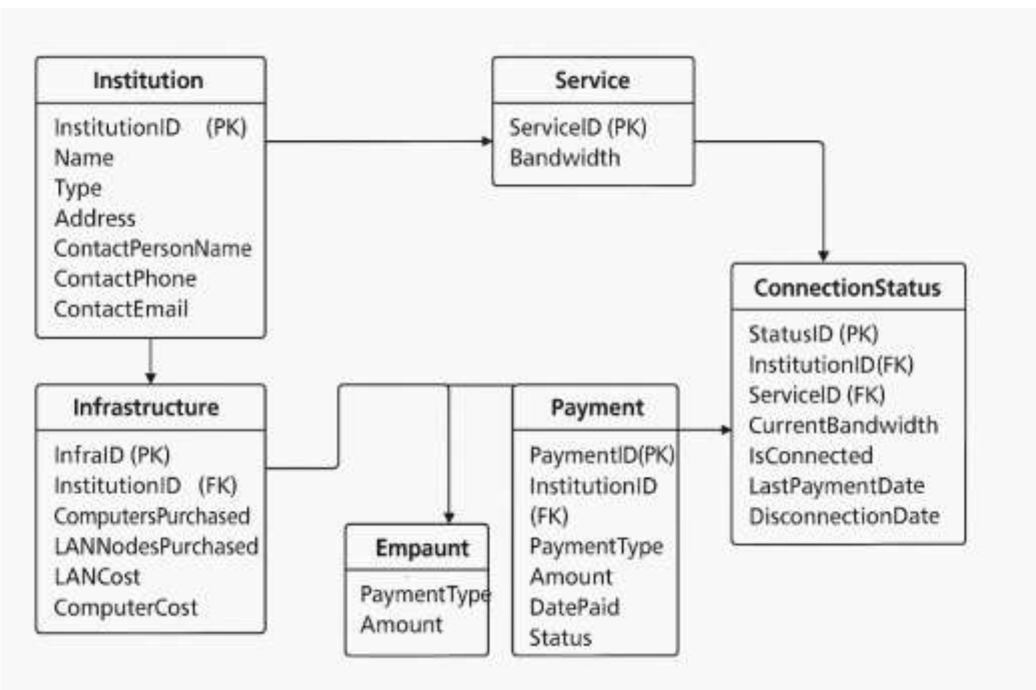
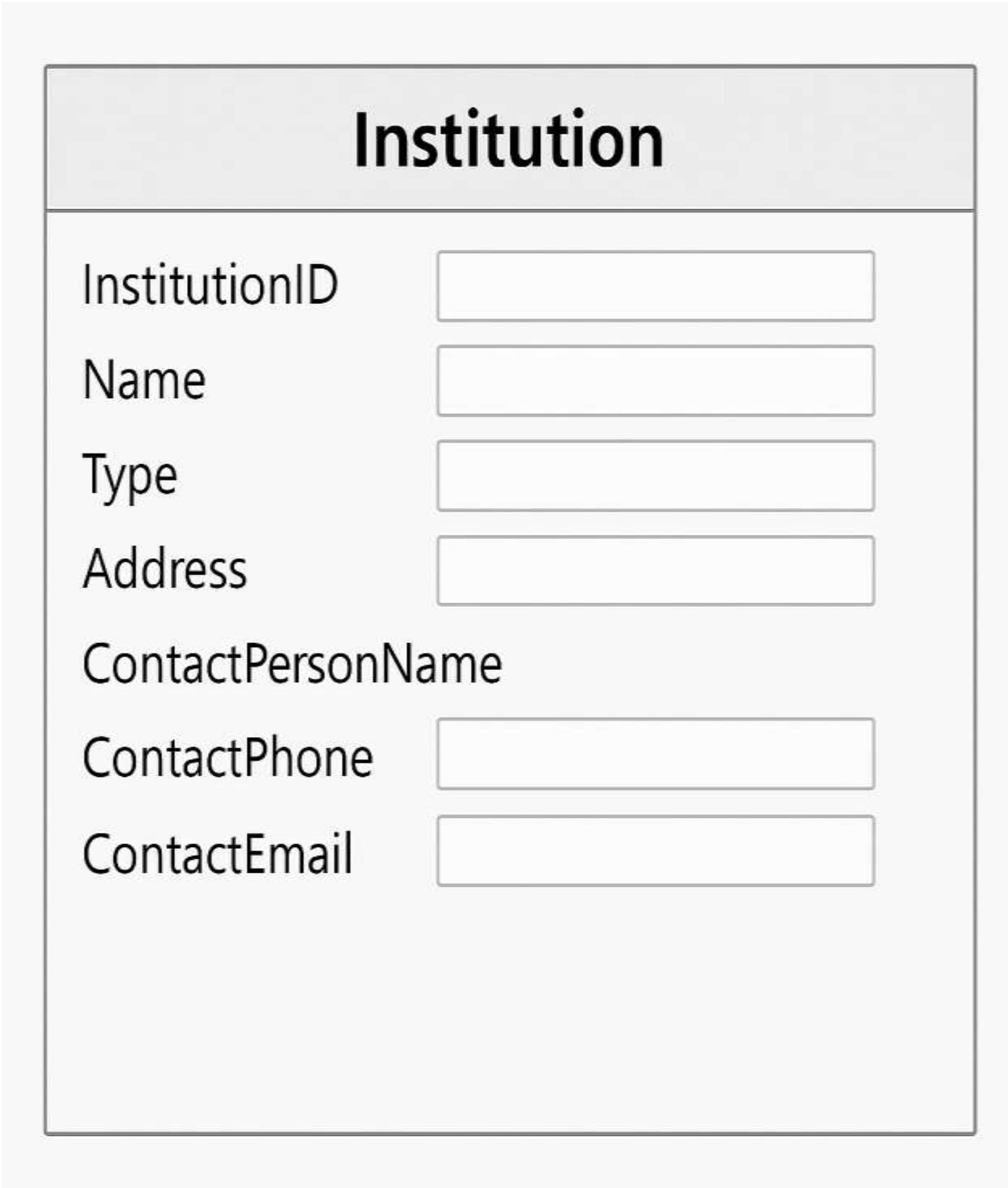
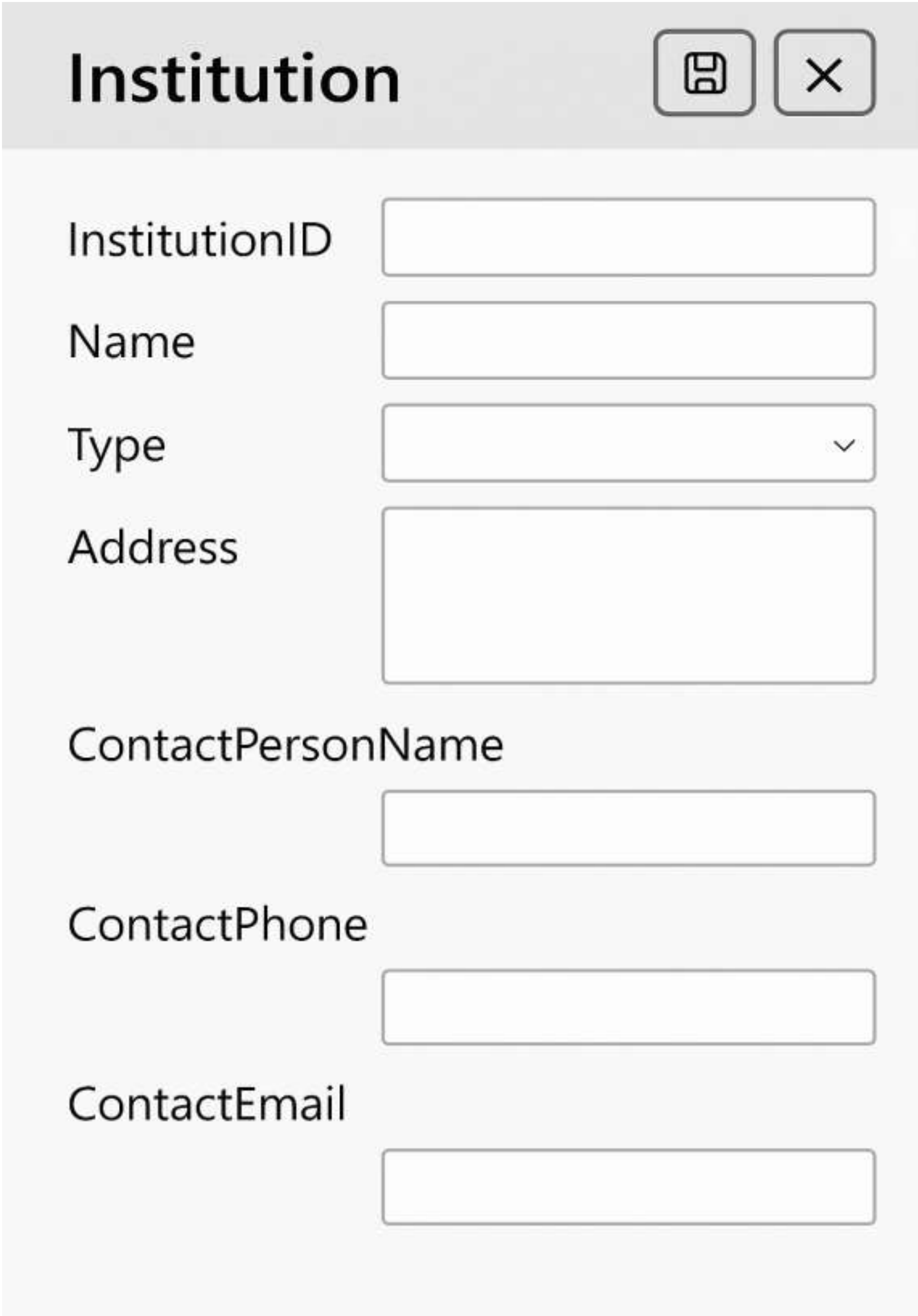
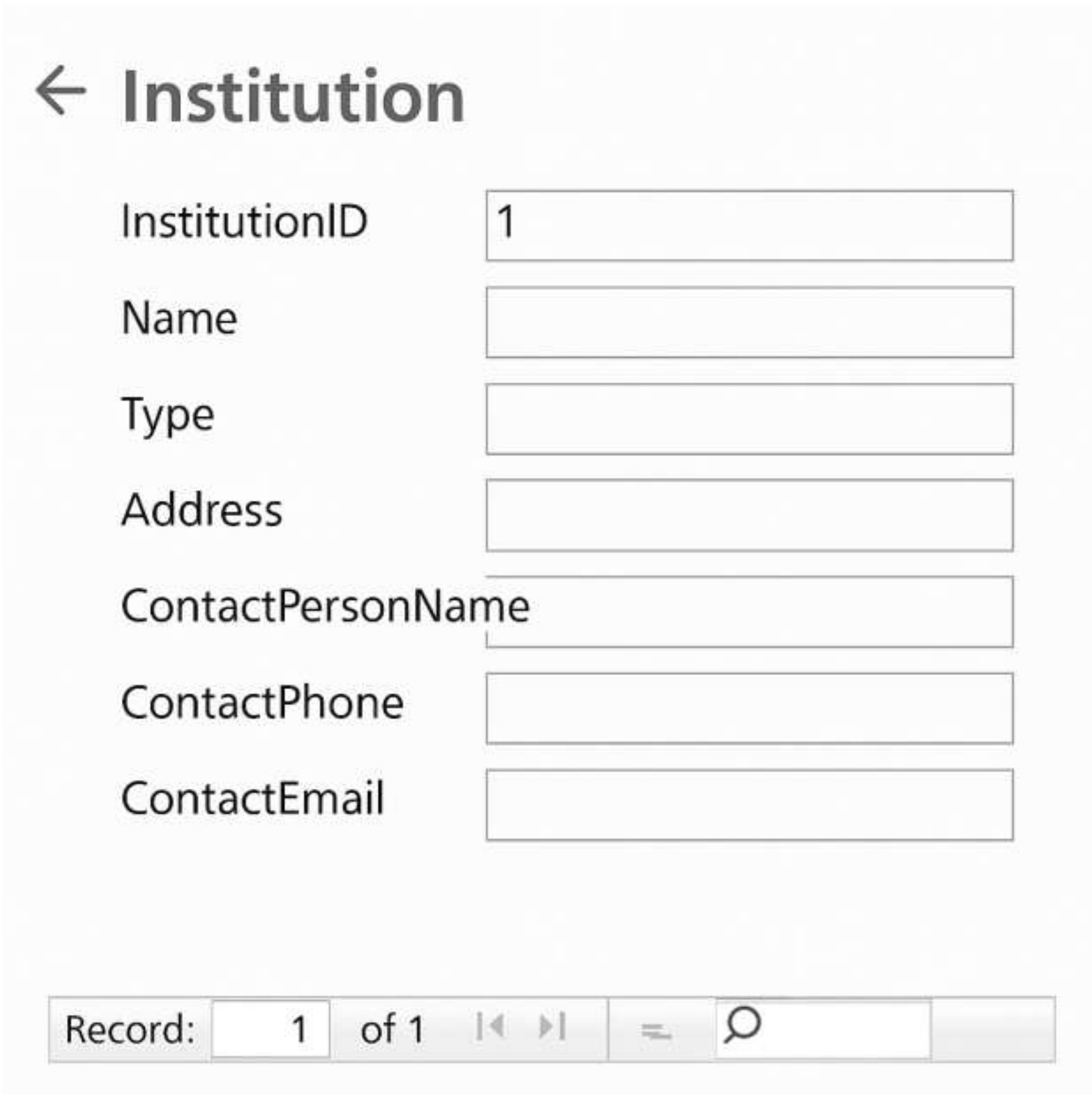


Figure 3 Entity Relation Diagram


Institution
InstitutionID
Name
Type
Address
ContactPersonName
ContactPhone
ContactEmail


Institution
InstitutionID
Name
Type
Address
ContactPersonName
ContactPhone
ContactEmail


Institution
InstitutionID
1
Name
Type
Address
ContactPersonName
ContactPhone
ContactEmail
Record: 1 of 1

Output Designs

Customer List Report		
Customer ID	Name	Contact Number

Invoice	
Invoice ID: Date.	Amount
Service	**Amount**
Internet Package	10-90
	Total

Payment Summary Report		
Payment ID.	Customer ID	Amount Paid
P801	101	
P002	102	
P900	103	

- Headers and triles
- Tables with gridlines
- Summary fields